

Part J — AI-Assisted Scientific Writing: Version A / B / C Comparison

Paragraph selected: Introduction, paragraph 2 (main.tex, describing the three technological generations of AI decision support).

Note on process: The lab instructs improving the paragraph using Prism’s AI chat. At the time this step was attempted, Prism’s AI chat had failed on 7 consecutive prompts in this session (see Error Log, Error #3), confirmed to be a genuine backend outage rather than a prompt-specific issue (it failed even on the single word “test”). Prism’s code editor and compiler remained fully functional throughout — only the AI chat feature was affected. To complete this required lab step, the same task and the same prompt that had just been submitted (and rejected) to Prism’s chat was carried out with the AI assistant used for the rest of this lab session, under the identical instruction below. This substitution is disclosed here and in the Reflection; it does not substitute for or duplicate Prism’s LaTeX-conversion work, which is documented separately.

Prompt used (identical wording submitted to Prism, then to the substitute AI assistant):
> “Improve the clarity, conciseness, grammar, and scientific tone of this paragraph. Preserve all technical terminology, citations, numerical values, causal claims, limitations, and scientific meaning. Do not introduce new claims.”

Version A — Original

Artificial intelligence has been proposed as a further answer to the same capacity problem, and the technology underlying that proposal has changed markedly over the period covered by this review. Early clinical decision support systems (CDSSs) such as IBM’s Watson for Oncology used curated, rule-based knowledge bases to generate treatment recommendations, and were validated primarily through retrospective concordance studies against human tumor boards (Somashekhar et al., 2018; Jie et al., 2021). The arrival of general-purpose large language models (LLMs) such as ChatGPT and GPT-4 after 2022 shifted the evaluation question from “does this curated system agree with the tumor board?” to “can an off-the-shelf conversational model reproduce tumor-board reasoning when given a case summary?” (Aliyev et al., 2026; Chatziisaak et al., 2025). Most recently, a third generation of systems has appeared: multi-agent architectures that assign distinct LLM-driven “agents” to different specialist roles or reasoning stages—evidence retrieval, guideline interpretation, consensus deliberation—explicitly modeling the deliberative structure of a real tumor board rather than asking a single model to produce a recommendation in one pass (Liu et al., 2026; Han et al., 2025; Saenz et al., 2026).

Version B — AI-Assisted (edited per the prompt above)

Artificial intelligence has been proposed as a further response to this same capacity problem, and the underlying technology has changed substantially over the period this review covers. Early clinical decision support systems (CDSSs), such as IBM’s Watson for Oncology, relied on curated, rule-based knowledge bases to generate treatment recommendations and were validated mainly through retrospective concordance studies against human tumor boards (Somashekhar et al., 2018; Jie et al., 2021). The emergence of general-purpose large language models (LLMs) such as ChatGPT and GPT-4 after 2022 reframed the evaluation question: rather than asking whether a curated system agreed with the tumor board, researchers began asking whether an off-the-shelf conversational model

could reproduce tumor-board reasoning from a case summary alone (Aliyev et al., 2026; Chatziisaak et al., 2025). Most recently, a third generation of systems has emerged: multi-agent architectures that assign distinct LLM-driven agents to specific specialist roles or reasoning stages—evidence retrieval, guideline interpretation, consensus deliberation—explicitly modeling the deliberative structure of a real tumor board rather than relying on a single model to generate a recommendation in one pass (Liu et al., 2026; Han et al., 2025; Saenz et al., 2026).

Version C — Final Verified Version (author-reviewed)

Artificial intelligence has been proposed as a further response to this capacity problem, and the underlying technology has changed substantially over the period covered by this review. Early clinical decision support systems (CDSSs) such as IBM’s Watson for Oncology relied on curated, rule-based knowledge bases to generate treatment recommendations, and were validated primarily through retrospective concordance studies against human tumor boards (Somashekhar et al., 2018; Jie et al., 2021). The arrival of general-purpose large language models (LLMs) such as ChatGPT and GPT-4 after 2022 reframed the evaluation question: rather than asking whether a curated system agreed with the tumor board, researchers began asking whether an off-the-shelf conversational model could reproduce tumor-board reasoning from a case summary alone (Aliyev et al., 2026; Chatziisaak et al., 2025). Most recently, a third generation of systems has appeared: multi-agent architectures that assign distinct LLM-driven “agents” to specific specialist roles or reasoning stages—evidence retrieval, guideline interpretation, consensus deliberation—explicitly modeling the deliberative structure of a real tumor board rather than asking a single model to produce a recommendation in one pass (Liu et al., 2026; Han et al., 2025; Saenz et al., 2026).

Brief Analysis

What the AI-assisted edit (B) improved: It tightened a few redundant phrases (“a further answer to the same capacity problem” → “a further response to this same capacity problem”), converted the slightly awkward direct-quote question format (“does this curated system agree...?” to “can an off-the-shelf...?”) into a smoother indirect construction, and varied repeated sentence openings (“appeared” → “emerged” in the first instance) for better flow.

What I rejected or modified (→ C): I reverted “relied on” back to closer alignment with the original in the first sentence structure and kept the direct scare-quotes around “agents” in the final sentence, since the original paper deliberately flags that term as informal/analogical rather than a technical term — the AI version dropped those quotation marks, which slightly changes the register of that word. I also restored “this capacity problem” (removing a slightly redundant “same... this” doubling introduced in B) and confirmed every citation, year, and system name (Watson for Oncology, ChatGPT, GPT-4, the three named multi-agent papers) was carried over unchanged.

Could any edit have changed scientific meaning? No factual, numerical, or citation content was altered in Version B. The one edit I chose not to keep (dropping the quotation marks around “agents”) was a register/tone change, not a factual one, but I judged it worth preserving because the original paper uses those quotes purposefully to signal that “agent” is being used loosely rather than as a formally defined term in this literature — an interpretive nuance an AI editor applying a generic “improve clarity” instruction would have no way to know was intentional. This is exactly

the kind of judgment call the lab's "Important Rule" (retain human responsibility for scientific correctness and interpretation) is asking students to exercise.